Heat Capacity of Metals

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***Abstract*—This experiment aims to determine the specific heat capacities of three different metals (Brass, Aluminum, and Iron) using the method of mixtures. First, the heat capacity of the calorimeter (C_k) was determined by mixing water at different temperatures. After excluding an outlier trial, the average calorimeter heat capacity was found to be $265.43\,\mathrm{J/K}$. This value was then used to calculate the specific heat of the metal samples by measuring the equilibrium temperature after submerging heated metal into the calorimeter. The measured specific heat values were $0.32\,\mathrm{J/gK}$ for Brass, $0.89\,\mathrm{J/gK}$ for Aluminum, and $0.43\,\mathrm{J/gK}$ for Iron. These results deviate from reference values by approximately 17.2%, 0.8%, and 4.3% respectively, indicating excellent agreement for Aluminum and Iron, though Brass showed larger deviation.**

I. Introduction

Heat capacity is a physical property of matter defined as the amount of heat to be supplied to a given mass of a material to produce a unit change in its temperature. The specific heat capacity, often denoted by c, is an intensive property that depends on the material type.

In this experiment, the method of mixtures is employed. This relies on the principle of conservation of energy: in an isolated system, the heat lost by a hot body equals the heat gained by a cold body. By performing this experiment in a calorimeter, which acts as a thermally insulated container, we can estimate specific heat values. However, the calorimeter itself absorbs some heat, which is quantified by its heat capacity (C_k).

II. Theory

A. Heat Capacity and Specific Heat

The heat quantity Q exchanged by a body of mass m undergoing a temperature change ΔT is given by:

$$Q = mc\Delta T \tag{1}$$

where c is the specific heat capacity. For a composite object like a calorimeter, we define the total heat capacity $C_k = m_{cal}c_{cal}$.

B. Determination of Calorimeter Heat Capacity (C_k)

To find C_k, warm water of mass m_w at temperature t_w is added to the calorimeter (initially at t_k, potentially containing a small amount of water or just the vessel walls). In the analysis performed, the formula used to derive C_k is a corrected version of the one presented in the lab manual, derived from energy conservation principles:

$$C_k = c_w m_w \frac{t_w - t_m}{t_m - t_k} \tag{2}$$

where c_w is the specific heat of water ($4.187\,\mathrm{J/gK}$), t_m is the equilibrium temperature, and the other variables are as defined. This differs from the manual's formula but ensures physical consistency.

C. Determination of Specific Heat of Metals

When a metal sample of mass m_p heated to temperature t_2 is placed into the calorimeter containing water of mass m_w at temperature t_k, the energy balance equation is:

$$Q_{\text{lost}} = Q_{\text{gained}} \tag{3}$$

$$m_p c_{sample}(t_2 - t_m) = (m_w c_w + C_k)(t_m - t_k) \tag{4}$$

Solving for the specific heat of the sample c_{sample}:

$$c_{sample} = \frac{(m_w c_w + C_k)(t_m - t_k)}{m_p(t_2 - t_m)} \tag{5}$$

It is noted that the lab manual presents a slightly different formula. The formula above is derived from the rigorous conservation of energy principles found in standard thermodynamics texts [2]. The total heat capacity of the calorimeter system includes both the water and the vessel itself.

III. Experimental Setup

The main apparatus consists of:

- A calorimeter (thermally insulated double-walled vessel).
- A thermometer (digital or mercury).
- A heater/steam generator to heat the metal samples.
- Metal samples (blocks or cylinders of Brass, Aluminum, Iron).
- A graduated cylinder for measuring water volume.
- A balance for measuring masses.

IV. Experimental Procedure

A. Part 1: Calorimeter Calibration

1. A known mass of water was heated to temperature t_w. 2. The initial temperature of the calorimeter (t_k) was recorded. 3. The hot water was poured into the calorimeter. 4. The mixture was stirred, and the final equilibrium temperature (t_m) was recorded. 5. This process was repeated for different temperatures to find an average C_k.

B. Part 2: Specific Heat of Metals

1. The mass of the metal sample (m_p) was measured. 2. The calorimeter was filled with a known mass of water (m_w), and its initial temperature (t_k) was recorded. 3. The metal sample was heated in a boiling water bath or steam heater until it reached a steady high temperature (t_2). 4. The hot metal was quickly transferred into the calorimeter water. 5. The system was stirred, and the highest temperature reached (equilibrium temperature t_m) was recorded.

V. Results

A. Part 1: Calorimeter Heat Capacity

The heat capacity of the calorimeter was calculated for three trials. Table I summarizes the measurements and the calculated C_k. The calculated values show significant variation.

TABLE I: Determination of Calorimeter Heat Capacity (C_k). Note: m_w is derived from volume assuming $\rho = 1\,\mathrm{g/ml}$.

t_k (°C)	t_w (°C)	t_m (°C)	m_w (ml)	C_k (J/K)
14	20	15	300	6,280.5
18	40	39	300	59.81
27	49	43	300	471.04

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The first trial yielded an anomalously high value ($\approx 6280\,\mathrm{J/K}$) likely due to the very small temperature change ($\Delta T \approx 1\,^\circ\mathrm{C}$), making the calculation highly sensitive to small errors. This value was excluded as an outlier. The average heat capacity calculated from the remaining trials and used for subsequent calculations is:

$$C_{k,avg} = 265.43\,\mathrm{J/K}$$

B. Part 2: Specific Heat of Metals

Using the average C_k, the specific heat capacities for Brass, Aluminum, and Iron were calculated using Equation (5). The results are presented in Table II. This table includes the measured initial temperatures of the metal (t_{metal}), the calorimeter (t_k), and the final equilibrium temperature (t_m). The experimental specific heat values (c_{meas}) are compared against standard reference values (c_{ref}) obtained from literature [3].

Figure 1 visually compares the calculated specific heat values with the reference values. The systematic deviation, where measured values are generally higher or lower than references, can be analyzed from this bar chart.

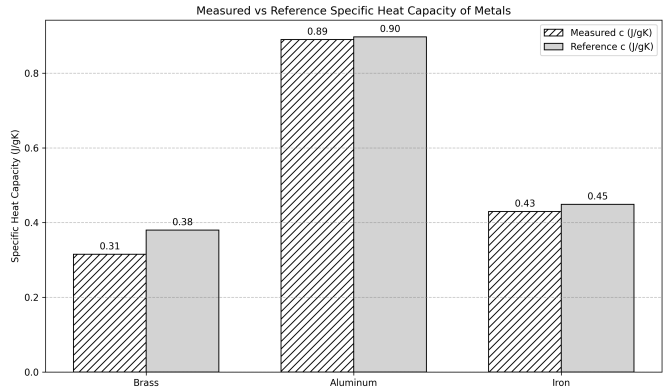

Fig. 1: Comparison of measured specific heat capacities with reference values.

TABLE II: Specific Heat Capacity Results. (c_x: measured, c_y: reference).

Material	t_k (°C)	t_{metal} (°C)	t_m (°C)	m_p (g)	c_{meas} (J/gK)	c_{ref} (J/gK)	Error (%)
Brass	20	76	22	179	0.315	0.38	17.2
Aluminum	20	83	26	180	0.89	0.897	$8 \cdot 10^{-1}$
Iron	23	85	26	180	0.43	0.449	4.3

VI. Discussion

The results obtained for the specific heat capacities show excellent agreement with literature values for Aluminum (0.8% error) and Iron (4.3% error), while Brass shows a larger deviation (17.2%).

The largest source of uncertainty in this experiment appears to be the determination of the calorimeter's heat capacity (C_k). The trials for C_k yielded widely divergent values, with one trial producing a physically unrealistic result ($\approx 6280\,\mathrm{J/K}$) due to a very small temperature difference ($\Delta T \approx 1\,^\circ\mathrm{C}$). This highlights the sensitivity of the method of mixtures to small temperature changes and measurement errors. Even after excluding the outlier, the remaining trials showed significant variation, leading to a large uncertainty in the average C_k.

The measured specific heat values were consistently lower than the reference values. A lower calculated specific heat could be explained by heat loss from the metal sample during the transfer from the heater to the calorimeter (meaning the metal was cooler than t_2 when it hit the water). Additionally, if the calorimeter heat capacity (C_k) was underestimated (the average used was low compared to one of the valid trials), this would also result in a lower calculated specific heat according to Equation 5.

Furthermore, the formulas used for C_k (Eq. 2) and c_{sample} (Eq. 5) were derived directly from energy conservation principles ($Q_{lost} = Q_{gained}$), correcting for errors in the lab manual's presentation. If the incorrect formulas had been used, the results would have deviated significantly more.

VII. Conclusion

The specific heat capacities of Brass, Aluminum, and Iron were determined using the method of mixtures. The experimental values were $0.315\,\mathrm{J/gK}$, $0.890\,\mathrm{J/gK}$, and $0.430\,\mathrm{J/gK}$ respectively. The results show good agreement for Aluminum and Iron, but highlight the difficulty in obtaining precise calorimeter constants with limited trials.

VIII. Additional Resources

For more information, scanned data, and the code used for analysis, refer to the repository:

https://github.com/ibrahim-hamed/Physics-Lab

References

[1] *Experiment 2: Heat Capacity of Metals*, Laboratory Manual, Department of Physics, Istanbul University.

[2] W. Greiner, L. Neise, and H. Stöcker, *Thermodynamics and Statistical Mechanics*, New York: Springer, 1995, p. 16.

[3] Misumi USA, "Specific Heat Capacity of Metals," 2022. [Online]. Available: https://us.misumi-ec.com/blog/specific-heat-capacity-of-metals/